

304/304L Stainless Steel

UNS S30400/S30403

Description

304

STAINLESS STEEL

304 stainless steel is also known as 18/8 stainless steel, meaning that the steel containing more than 18% chromium and more than 8% nickel. The alloy is a one of the most common stainless steel materials in the market. 304 stainless steel has a wide range of applications with good corrosion resistance, heat resistance, low temperature strength and mechanical properties, good stamping, bending and hot processing. It also has no heat treatment hardening phenomenon under the operating temperature of -196 °C to 800 °C. 304L stainless steel is a low carbon version of 304 stainless steel. Through the controlled addition of nitrogen, it is common for 304L to meet the mechanical properties of 304 straight grade. As a result, most products are dual certified as 304 and 304/304L.

Chemical Component

Ni	Cr	Mn	Si	C	S	P	N	Fe
8.0-10.5%	18.0-20.0%	2.0% max	0.75% max	0.03% max	0.03% max	0.045% max	0.10% max	Balance

Basic Information

Items	304/304L Stainless Steel
Catalog	ACMA-Fe-A008
Density	0.285 lb/in ³
Electrical Resistivity, Ohm-circ mil/ft	433
Poisson's Ratio	0.3

Items**304/304L Stainless Steel**

Melting Range

2550-2590 °F

Coefficient of Thermal
Expansion, in/in°F x 10⁻⁶

70-212 °F: 9.2

Thermal Conductivity,
Btu·ft/ft²·hr·°F

At 212 °F: 9.4; At 392 °F: 12.4

Modulus of Elasticity Dynamic,
psi x 10⁶

At 70 °F: 29

Material Forms

Alfa Chemistry offers 304/304L stainless steel in a variety of forms, including **Concentric Reducers, Fittings, Flanges, Hex Bar, Pipe, Plate, Sheet, Stub Ends, Tees, Tubing, Wire, Strip.**

Mechanical Properties

- **Representative tensile properties**

Ultimate Tensile Strength	0.2% Yield Strength	Elongation	Hardness MAX
75 ksi	30 ksi	40%	201 Brinell

- **Typical tensile and impact properties**

Temperature	Ultimate Tensile Strength	0.2% Yield Strength	HCharpy Impact V-notch
-425 °F	250 ksi	100 ksi	85-320 ft-lbs
-320 °F	230 ksi	70 ksi	85 ft-lbs
-100 °F	150 ksi	50 ksi	-
70 °F	90 ksi	35 ksi	150 ft-lbs

Temperature	Ultimate Tensile Strength	0.2% Yield Strength	HCharpy Impact V-notch
400 °F	70 ksi	23 ksi	-
800 °F	66 ksi	19 ksi	-
1200 °F	48 ksi	15.5 ksi	-
1500 °F	23 ksi	23 ksi	-

Specifications

W. NR.: 1.4301, 1.4307

ASME: SA 240, SA 276, SA 312, SA 479

ASTM: A 240, A 276, A 312, A 479

AMS: 5511, 5513, 5639, 5647

Features

- Good general corrosion resistance
- Ease of cleaning
- Excellent strength and toughness at cryogenic temperatures
- Good formability
- Good weldability

Applications

- Food processing and handling
- Heat exchangers
- Chemical process vessels
- Conveyors
- Architectural